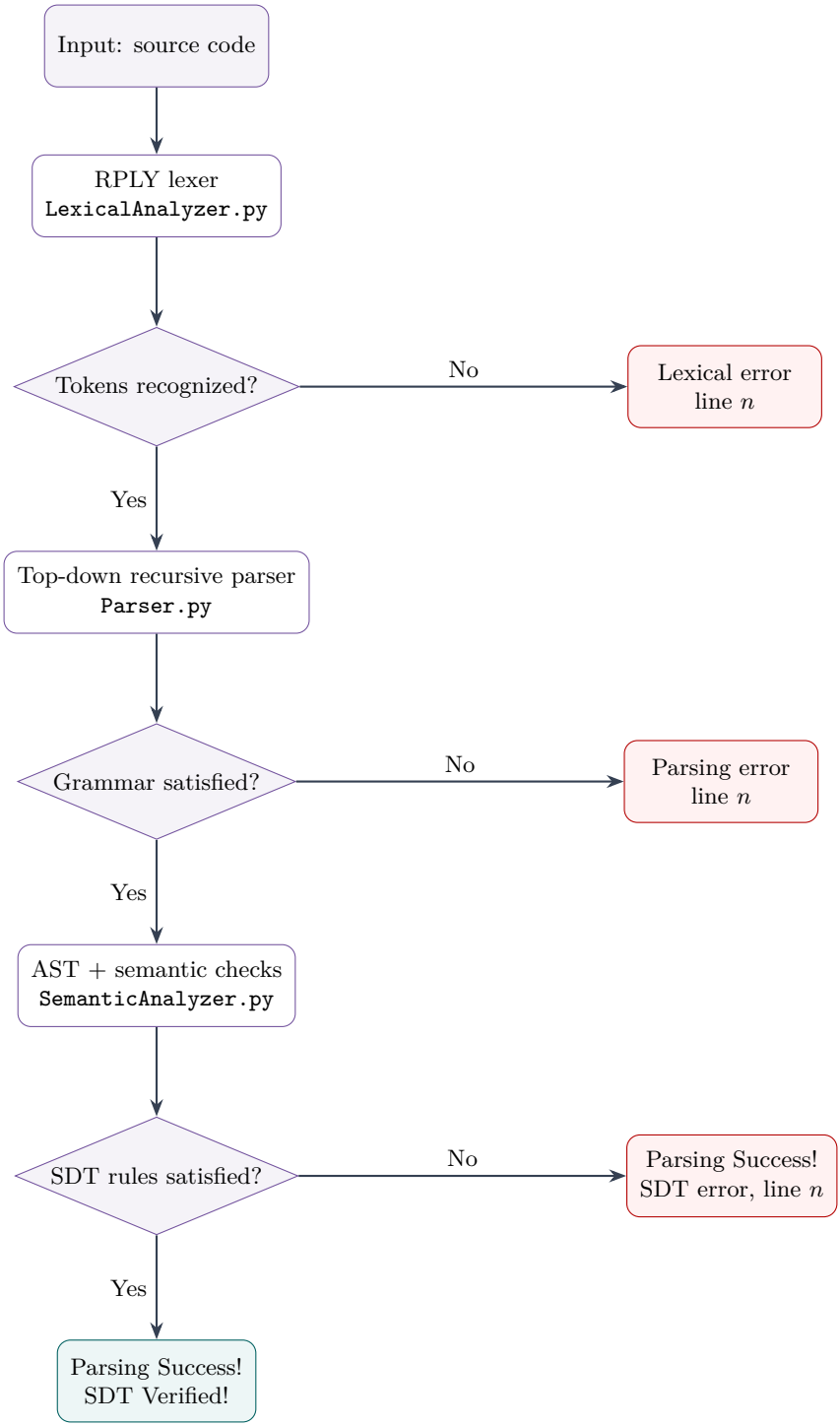

Input: source code
RPLY lexer
LexicalAnalyzer.py
Tokens recognized?
No
Lexical error
line $n$
Yes
Top-down recursive parser
Parser.py
Grammar satisfied?
No
Parsing error
line $n$
Yes
AST + semantic checks
SemanticAnalyzer.py
SDT rules satisfied?
No
Parsing Success!
SDT error, line $n$
Yes
Parsing Success!
SDT Verified!